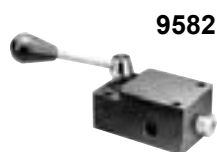
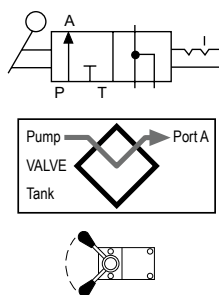


**3-WAY / 2-POSITION MANUAL VALVES**

**9582**

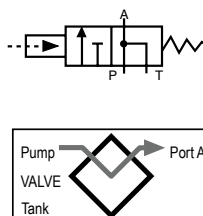
**9584**

**Applications:** Single-acting cylinders.

**Actuation:** Lever-operated.

**Functions:** Cylinder piston "advance," "hold" and "return."

**Used on these pumps:** P460, PE17, PE21, PE30, PE46, PE55, and PE120 series.

| Order No. | Description                                                   |
|-----------|---------------------------------------------------------------|
| 9582      | 3-Way/2-Position manual valve. Wt., 1,1 kg.                   |
| 9584      | 3-Way/2-Position manual valve "flipper" control. Wt., 0,8 kg. |

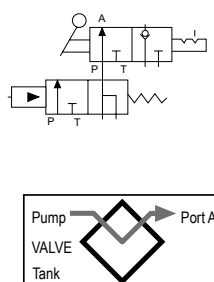
**3-WAY / 2-POSITION PILOT-OPERATED AUTOMATIC VALVES**

**Applications:** Single-acting cylinders.

**Actuation:** Pilot oil.

**Functions:** When pump is started, pilot oil automatically closes valve and directs oil to cylinder. When pump is stopped, valve automatically opens and oil returns to reservoir.

**Used on these pumps:** Furnished with pilot lines and adapters for PA55, PA90, PE30, PE55, and PE120 series.

| Order No. | Description                                                   |
|-----------|---------------------------------------------------------------|
| 9610      | 3-way/2-position pilot-operated automatic valve. Wt., 1,9 kg. |

**2/3-WAY/2-POSITION MANUAL/PILOT-OPERATED AUTOMATIC VALVE**

**Applications:** Manual operation for load lifting and holding with single-acting cylinders; automatic "dump" for operating hydraulic tools.

**Actuation:** Flipper lever/pilot oil.

**Functions:** With lever in closed position, valve will hold the load. When lever is "open," valve functions as a true automatic "dump" valve..

**Used on these pumps:** Furnished with pilot lines and adapters for PA55, PA90, PE30, PE55, and PE120 series. For application on other pumps, consult factory.

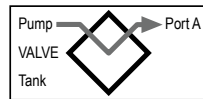
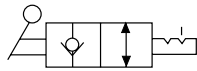
| Order No. | Description                                                          |
|-----------|----------------------------------------------------------------------|
| 9610A     | 2/3-way/2-position manual/pilot-operated automatic valve. Wt., 2 kg. |

**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

**IMPORTANT:** Conversion kit 251528 must be used when mounting any of the valves on this page on PA17 or PE17 pumps.

**IMPORTANT:** When ordering any valve for a PE30 or PG30 series pump, 12,7 mm longer mounting screws are required. For valves 9504, 9584, 9610 and 9610A, order four 12001 cap screws. For valve 9582, order two 12001 and two 10856 cap screws.

## 2-WAY/2-POSITION MANUAL VALVE



**Applications:** Single-acting cylinders.

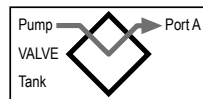
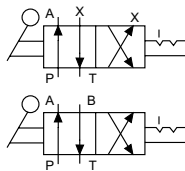
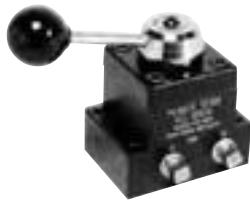
**Actuation:** Flipper lever-operated.

**Functions:** Cylinder piston "advance," "hold" and "retract."

**Used on these pumps:** PE172 and PA172.

| Order No. | Description                                 |
|-----------|---------------------------------------------|
| 9517      | 2-way/2-position manual valve. Wt., 1,5 kg. |

## 3/4-WAY/2-POSITION MANUAL VALVE



**Applications:** Single or double-acting cylinders.

**Actuation:** Lever-operated, detent-positioned.

**Functions:** Pos. 1 – Oil is directed to "advance" side of cylinder, oil from "retract" side goes to reservoir, cylinder "holds" with pump shut-off. Pos. 2 – Oil goes to "retract" side of cylinder, cylinder "holds" with pump shut-off. When using as a 3-way valve for single-acting cylinders, port "A" or "B" is plugged. See note below regarding plugging of ports and resulting heat build-up..

**Used on these pumps:** P460, PA6D, PA17, PA46, PA55, PA60, PE17, PE21, PE30, PE46, PE55, PE120, PE400, PQ60 and PQ120 series.

| Order No. | Description                                   |
|-----------|-----------------------------------------------|
| 9504      | 3/4-way/2-position manual valve. Wt., 1,9 kg. |

**NOTE:** 9504 can be remote-mounted with a 9510 sub-plate.

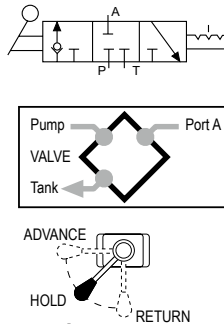
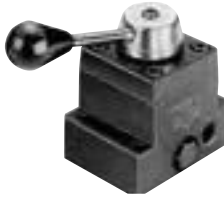
**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

**IMPORTANT:** Conversion kit 251528 must be used when mounting any of the valves on this page on PA17 or PE17 pumps.

**IMPORTANT:** When ordering any valve for a PE30 or PG30 series pump, 12,7 mm longer mounting screws are required. For valves 9504, 9584, 9610 and 9610A, order four 12001 cap screws. For valve 9582, order two 12001 and two 10856 cap screws.

**NOTE:** Valves 9501, 9502, 9504 and 9507 can have a port blocked or have a closed center position. When a port is blocked and the valve is shifted to the blocked port, the pump will generate excessive heat. An electric or rotary air pump can either be turned off manually or with a pressure switch. Reciprocating air pumps may be adjusted to stall out and stop.

## 3-WAY/3-POSITION (CLOSED CENTER) NON-INTER FLOW MANUAL VALVE WITH POSI-CHECK®



**Applications:** Single-acting cylinders.

**Actuation:** Lever-operated, detent-positioned.

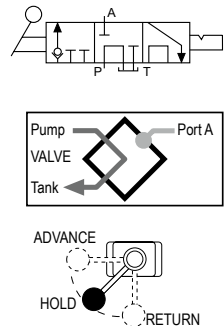
**Functions:** Pos. 1 – Oil is directed from pump to cylinder and “holds” with pump shut-off, line to reservoir is blocked. Pos. 2 – All oil is open to reservoir through tank line. Center Pos. – Cylinder pressure is held, and pump should be shut-off.

**Used on these pumps:** P460, PA17, PA46, PA55, PA60, PE17, PE21, PE30, PE46, PE55, PE120, PE400, PQ60 and PQ120 series.

| Order No. | Description                                                 |
|-----------|-------------------------------------------------------------|
| 9502      | 3-way/3-position (closed center) manual valve. Wt., 1,9 kg. |

**NOTE:** A pressure switch and/or gauge may be attached if desired. Also, the 9502 can be remote-mounted if a 9510 sub-plate is used.

## 3-WAY/3-POSITION (TANDEM CENTER) MANUAL VALVE WITH POSI-CHECK® VALVE



**Applications:** Single-acting cylinders.

**Actuation:** Lever-operated, detent-positioned.

**Functions:** “Advance,” “hold” and “return.” When shifted to “return” position, pump and cylinder return oil through their own separate return lines, allowing faster retraction of piston. The Posi-Check® feature guards against pressure loss when shifting from “advance” to “hold” position.

**Used on these pumps:** P460, PA17, PA46, PA55, PE17, PE21, PE30, PE46, PE55, PE120, PQ60, PQ120, PE400, PG30, PG55, PG120 and PG400 series.

| Order No. | Description                                                 |
|-----------|-------------------------------------------------------------|
| 9520      | 3-way/3-position (tandem center) manual valve. Wt., 2,3 kg. |

**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

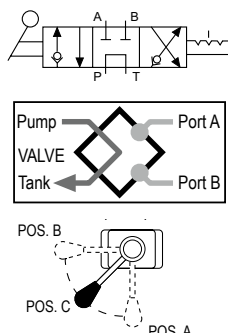
**NOTE:** Valves 9501, 9502, 9504 and 9507 can have a port blocked or have a closed center position. When a port is blocked and the valve is shifted to the blocked port, the pump will generate excessive heat. An electric or rotary air pump can either be turned off manually or with a pressure switch. Reciprocating air pumps may be adjusted to stall out and stop.

**NOTE:** Gauge ports monitor pump pressure only, not pressure to the hydraulic cylinder(s).

**IMPORTANT:** Conversion kit 251528 must be used when mounting any of the valves on this page on PA17 or PE17 pumps.

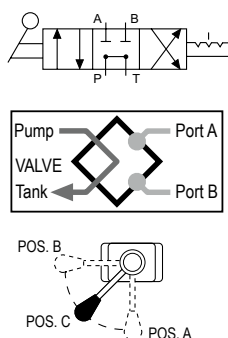
**IMPORTANT:** When ordering any valve for a PE30 or PG30 series pump, 12,7 mm longer mounting screws are required. For valves 9502 and 9520, order four 12001 cap screws. For valve 9576, order four 17428 cap screws.

## 4-WAY/3-POSITION (TANDEM CENTER) VALVE WITH POSI-CHECK®

**Applications:** Single or double-acting cylinders.**Actuation:** Lever-operated, detent-positioned.**Functions:** "Advance," "hold" and "return." The Posi-Check® feature guards against pressure loss when shifting from "advance" to "hold" position.**Used on these pumps:** P460, PA6D, PA17, PA46, PA55, PE17, PE21, PE30, PE46, PE55, PE84, PE120, PE400, PED, PG30, PG55, PG120, PG400, PQ60 and PQ120 series.

| Order No. | Description                                                 |
|-----------|-------------------------------------------------------------|
| 9506      | 4-way/3-position (tandem center) manual valve. Wt., 2,3 kg. |

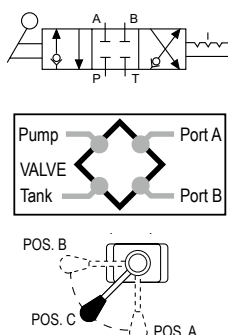
## 4-WAY/3-POSITION (TANDEM CENTER) AND (OPEN-CENTER) MANUAL VALVES

**Applications:** Single or double-acting cylinders.**Actuation:** Lever-operated, detent-positioned.**Functions:** The 9500 provides "advance," "hold" and "return." The 9511 (open center) valve can be used if holding is not a requirement, as when running two separate hydraulic tools. Provides "advance" and "return" only.**Used on these pumps:** P460, PA17, PA46, PA55, PE17\*, PE21, PE30, PE46, PE55, PE120, PE400, PG30, PG55, PG120, PG400, PQ60 and PQ120 series.

\*Does not mount without 251528.

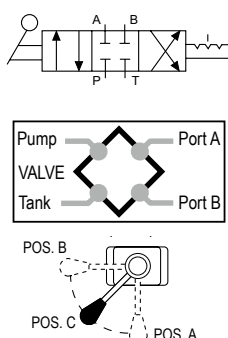
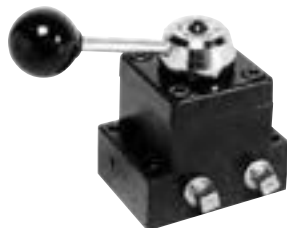
| Order No. | Description                                                 |
|-----------|-------------------------------------------------------------|
| 9500      | 4-way/3-position (tandem center) manual valve. Wt., 1,9 kg. |
| 9511      | 4-way/3-position (open center) manual valve. Wt., 1,9 kg.   |

## 4-WAY/3-POSITION (CLOSED CENTER) MANUAL VALVE WITH POSI-CHECK®

**Applications:** Single or double-acting cylinders.**Actuation:** Lever-operated, detent-positioned.**Functions:** Similar to 9506, but is a closed center valve with Posi-Check®. Generally used to operate multiple cylinders with a single pump. Provides "advance," "hold" and "return." The Posi-Check® feature guards against pressure loss when shifting from the "advance" to "hold" position. See note below regarding plugging of ports and resulting heat build-up.**Used on these pumps:** P460, PA17, PA46, PA55, PA60, PA6D, PE17, PE21, PE30, PE46, PE55, PE120, PE400, PQ60 and PQ120 series.

| Order No. | Description                                                 |
|-----------|-------------------------------------------------------------|
| 9507      | 4-way/3-position (closed center) manual valve. Wt., 2,3 kg. |

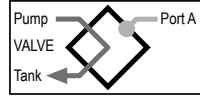
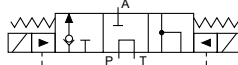
## 4-WAY/3-POSITION (CLOSED CENTER) MANUAL VALVE

**Applications:** Single or double-acting cylinders.**Actuation:** Lever-operated, detent-positioned.**Functions:** "Advance," "hold" and "return." Closed center design makes valve suitable for operating multiple cylinders from a single pump. See note below regarding plugging of ports and resulting heat build-up.**Used on these pumps:** P460, PA17, PA46, PA55, PA60, PE17, PE21, PE30, PE46, PE55, PE120, PE400, PQ60 and P120 series.

| Order No. | Description                                          |
|-----------|------------------------------------------------------|
| 9501      | 4-way/3-position (closed center) valve. Wt., 1,9 kg. |

**NOTE:** All valves on this page may be remote-mounted with a 9510 sub-plate. Valves 9501, 9502, 9504 and 9507 can have a port blocked or have a closed center position. When a port is blocked and the valve is shifted to the blocked port, the pump will generate excessive heat. An electric or rotary air pump can either be turned off manually or with a pressure switch. Reciprocating air pumps may be adjusted to stall out and stop.

## 3-WAY/3-POSITION (TANDEM CENTER) SOLENOID VALVES WITH POSI-CHECK®



**Applications:** Single-acting cylinders.

**Actuation:** Solenoid-operated: 9605 is 115V, 50/60 Hz; 9599 is 24V, 50/60 Hz.

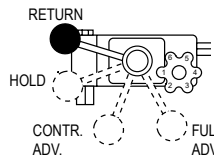
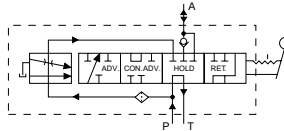
**Functions:** "Advance," "hold" and "return" positions. When in "advance," solenoid "B" is energized and oil goes from pump to cylinder through pressure port. In "return" position, solenoid "A" is energized and oil is directed from cylinder and pump to reservoir. With both solenoids de-energized, in "hold" position, oil from pump is directed back to reservoir while oil is checked in cylinder. The Posi-Check® feature holds load when shifting from "advance" to "hold" position.

**Used on these pumps:** Furnished with pilot lines and adapters for PE55, PE30 (carrying handles must be removed) and PE120 series. For application on other models, consult factory.

| Order No. | Description                                                                   |
|-----------|-------------------------------------------------------------------------------|
| 9605      | 3-way/3-position (tandem center) solenoid valve, 115V, 50/60 Hz. Wt., 6,4 kg. |
| 9599      | 3-way/3-position (tandem center) solenoid valve, 24V, 50/60 Hz. Wt., 6,4 kg.  |

**NOTE:** Valves above are shipped without controls. Use 202777 remote hand control. Consult factory for field installation.

## 3-WAY/4-POSITION MANUAL PRESSURE COMPENSATED VALVE



**Applications:** Single-acting cylinders. Primarily for use in testing soil, rock, concrete, asphalt and related engineering materials.

**Actuation:** Lever and adjustable, pressure compensated flow control valve.

**Functions:** Cylinder piston "return," "hold," "controlled advance" (pressure compensated) and "advance" (full flow). Will deliver a relatively constant flow regardless of pressure between 70 and 700 Bar.

**Used on these pumps:** PA17, PA46, PA55, PE17, PE21, PE30\*, PE46, PE55, PE200, PE400, PG30\*, PG55, PG120, PG400, PQ60 and PQ120 series. \* NOTE: Adapter kit 252161 is required for mounting this valve to a PE30 or PG30 series pump.

| Order No. | Description                                                    |
|-----------|----------------------------------------------------------------|
| 9609      | 3-way/4-position manual pressure compensated valve. Wt., 4 kg. |



## Pressure compensated valve performance chart

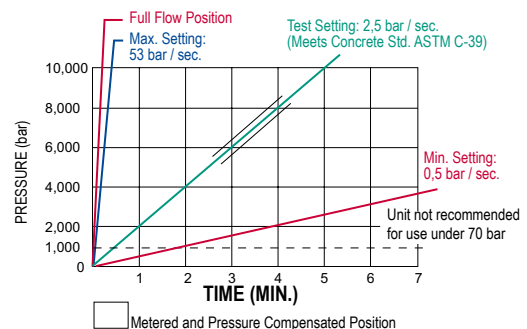
**Flow**

Full flow position - 19 l (Ref.)  
Metered advance position  
1 l/min. (Max.)

**Pressure**

Min. working pressure - 70 bar.  
Max. working pressure - 700 bar.  
Max. valve case pressure - 35 bar.

Chart to the right reflects valve performance when "metered advance" position is selected.

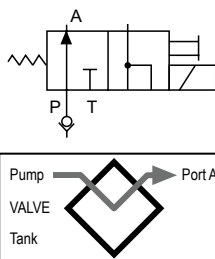


**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

**IMPORTANT:** Conversion kit 251528 must be used when mounting the 9609 valve on PA17 or PE17 pumps.

**IMPORTANT:** When ordering any valve for a PE30 or PG30 series pump, 12,7 mm longer mounting screws are required. For valves 9500, 9501 and 9511, order four 12001 cap screws. For valve 9552, 9506, and 9507, order four 11956 cap screws. For valves 9599 and 9605, order four 251078 cap screws. For valve 9609, order four 10855 cap screws.

## 3-WAY/2-POSITION SOLENOID VALVE



**Applications:** Single-acting cylinders.

**Actuation:** Solenoid-operated, 115V, 50/60 Hz.

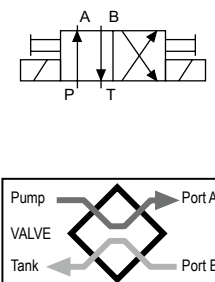
**Functions:** Cylinder piston advances when solenoid is de-energized and pump is running. When solenoid is energized, oil is directed to reservoir, and piston returns. For "hold" position, pump is stopped with solenoid de-energized.

**Used on these pumps:** PE17, PE21, PE30, PE46, PE55, PE120, PE400, PQ60 and PQ120 series.

| Order No. | Description                                                  |
|-----------|--------------------------------------------------------------|
| 9579      | 3-way/2-position solenoid valve, 115V, 50/60 Hz., Wt. 4,4 kg |
| 9569      | 3-way/2-position solenoid valve, 24V, 50/60 Hz., Wt. 4,4 kg  |
| 9570      | 3-way/2-position solenoid valve, 230V, 50/60 Hz., Wt. 4,4 kg |

**NOTE:** Valves above are shipped without control switch. Use 202777 remote hand switch. When this valve is mounted, the pump must be equipped with an outlet check valve.

## 3/4-WAY/2-POSITION SOLENOID VALVES



**Applications:** Single or double-acting cylinders. When used with single-acting cylinders, one port should be plugged.

**Actuation:** Solenoid-operated, 115V, 50/60 Hz.

**Functions:** Oil is directed to "extend" side of cylinder, oil from "retract" side goes to reservoir; cylinder "holds" with pump shut-off. Oil is directed to "retract" side of cylinder; oil from "extend" side goes to reservoir. **NOTE:** Cylinder will not "hold" in the "return" position with motor running or shut-off.

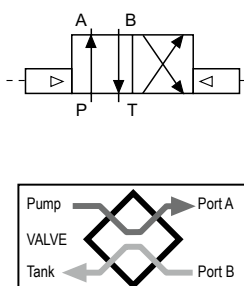
**Used on these pumps:** 9552, 9572 and 9592 are used with PE17, PE30 (with carrying handles removed), PE46, PE55, PE400, PQ60 and PQ120 series.

| Order No. | Description                                                    |
|-----------|----------------------------------------------------------------|
| 9592      | 3/4-way/2-position solenoid valve, 115V, 50/60 Hz., Wt. 6,6 kg |
| 9552      | 3/4-way/2-position solenoid valve, 230V, 50/60 Hz., Wt. 6,6 kg |
| 9572      | 3/4-way/2-position solenoid valve, 24V, 50/60 Hz., Wt. 6,6 kg  |

**NOTE:** Valves above are shipped without controls. The 9552, 9572 and 9592 can be used with the 304718 remote hand control.

**NOTE:** Ports are 1/4" NPTF.

## AIR ACTUATED VALVE



**Applications:** Single or double-acting cylinders. When used with single-acting cylinders, one port should be plugged.

**Actuation:** Air-operated.

**Functions:** Oil is directed to "extend" side of cylinder, oil from "retract" side goes to reservoir, cylinder "holds" with pump shut-off. Oil is directed to "retract" side of cylinder, oil from "extend" side goes to reservoir.

**NOTE:** Cylinder will not "hold" in the "return" position with motor running or shut-off.

**Used on these pumps:** PA17, PA46 and PA55 series.

| Order No. | Description                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------|
| 9594      | 3/4-way/2-position solenoid valve, air-operated (minimum of 4 bar air pressure required). Wt., 5 kg. |

**NOTE:** Valve above is shipped without controls. 9594 can be used with the 209593 remote hand control.

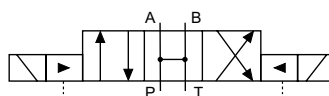
**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

**IMPORTANT:** Conversion kit 251528 must be used when mounting any of the valves on this page on PA17 or PE17 pumps.

**IMPORTANT:** When ordering any valve for a PE30 or PG30 series pump, 12,7 mm longer mounting screws are required. For valves 9569, 9570 and 9579, order four 10856 cap screws. For valves 9552, 9572 and 9592, order four 12001 cap screws.



## 4-WAY/3-POSITION (OPEN CENTER) SOLENOID VALVE



**Applications:** Double-acting cylinders.

**Actuation:** Solenoid-operated, 115V, 50/60 Hz.

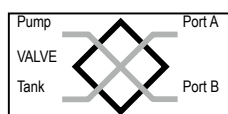
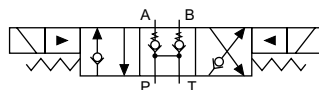
**Functions:** "Advance," open center and "return" positions. Cylinder ports and pump port are open to reservoir in "neutral."

**Used on these pumps:** Furnished with pilot lines and adapters for PE30 (with carrying handles removed), PE55 and PE120 series. For other pump models, consult factory.

**NOTE:** A pressure switch and/or gauge may be attached if desired.

| Order No. | Description                                                                   |
|-----------|-------------------------------------------------------------------------------|
| 9590      | 4-way/3-position (open center) solenoid valve, 115 volt, 50/60 Hz., Wt. 7 kg. |
| 9522      | 4-way/3-position (open center) solenoid valve, 230 volt, 50/60 Hz., Wt. 7 kg. |
| 9615      | 4-way/3-position (open center) solenoid valve, 24 volt, 50/60 Hz., Wt. 7 kg.  |

## 4-WAY/3-POSITION (TANDEM CENTER) PILOT-OPERATED SOLENOID VALVE



**Applications:** Double-acting cylinders.

**Actuation:** Solenoid-operated, 115V, 50/60 Hz.

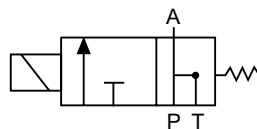
**Functions:** "Advance," "hold" and "return." The Posi-Check® feature holds the load when shifting from the "advance" to the "hold" position.

**Used on these pumps:** PE17, PE21, PE30 (with carrying handles removed), PE46, PE55, PE120, PE400, PQ60 and PQ120 series.

**NOTE:** A gauge may be attached if desired.

| Order No. | Description                                                                                                           |
|-----------|-----------------------------------------------------------------------------------------------------------------------|
| 9513      | 4-way/3-position (tandem center) solenoid valve, 115V, 50/60 Hz., Wt. 8.2 kg.                                         |
| 9512      | 4-way/3-position (tandem center) solenoid valve, 24V, 50/60 Hz., Wt. 8.2 kg.                                          |
| 9516      | 4-way/3-position (tandem center) solenoid valve, 12V, Wt. 8.2 kg. For use on the PG1204S and PG400 series pumps only. |
| 9519      | 4-way/3-position (tandem center) solenoid valve, 230V, 50/60 Hz., Wt. 8.2 kg. Consult factory for field installation. |

## 3-WAY/2-POSITION (PILOT-OPERATED, NORMALLY OPEN) SOLENOID VALVE



**Applications:** Single-acting cylinders.

**Actuation:** Solenoid-operated, 115V, 50/60 Hz.

**Functions:** "Advance" and "return."

**Used on these pumps:** Furnished with pilot lines and adapters for PE30 (with carrying handles removed), PE55 and PE120 series. For other pump models, consult factory.

**NOTE:** A pressure switch and/or gauge may be attached if desired.

| Order No. | Description                                                                    |
|-----------|--------------------------------------------------------------------------------|
| 9589      | 3-way/2-position (pilot-operated) solenoid valve, 115V, 50/60 Hz., Wt. 3.7 kg. |
| 9523      | 3-way/2-position (pilot-operated) solenoid valve, 230V, 50/60 Hz., Wt. 3.7 kg. |
| 9553      | 3-way/2-position (pilot-operated) solenoid valve, 24V, 50/60 Hz., Wt. 3.7 kg.  |

**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

**IMPORTANT:** Conversion kit 251528 must be used when mounting the 9609 valve on PA17 or PE17 pumps.

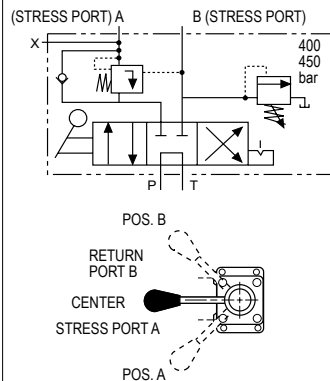
**IMPORTANT:** When ordering any valve for a PE30 or PG30 series pump, 12.7 mm longer mounting screws are required. For valves 9513 and 9519, order four 11956 cap screws. For valves 9523, 9553 and 9589, order four 10855 cap screws. For valves 9522, 9590 and 9615, order four 10854 cap screws.

**NOTE:** Valves above are shipped without control switch.

## 4-WAY/3-POSITION (TANDEM CENTER) MANUAL VALVE



Designed for use with Power Team air, gas-oil and electric powered hydraulic pumps.



**Applications:** Single strand, double-acting stressing jacks with Power Wedge seater.

**Actuation:** Lever-operated, detent-positioned.

**Operation:**

With valve in center position, pump is started.

Cable is inserted into stressing tool, valve is placed in "A" position. "Pull" portion of stressing tool is pressurized to specified level for proper cable tensioning ("A" port is checked internally, can only be released by building pressure in "B" position).

Valve is placed in "B" position, which is pressure controlled and will not exceed 440 bar. "Return" portion of stressing tool is pressurized and will release "A" port when pressure reaches approximately one-half the "A" port pressure. "A" port remains open as long as this pressure differential is maintained.

Pump is stopped, valve is placed in "A" position, releasing "B" port pressure.

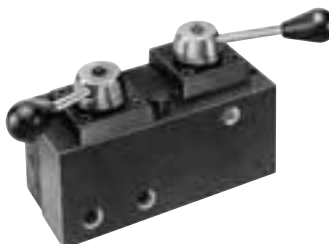
**Used on these pumps:** PA17\*, PA46\*, PA55, PE17\*, PE21\*, PE30, PE46\*, PE55, PE60, PE120, PE400, PG30\*, PG55, PG120, PG400, PQ60 and PQ120 series.

\* These pumps may have reduced first flow stage characteristics due to internal valve restrictions.

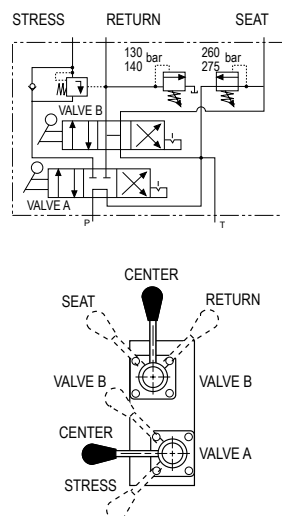
| Order No. | Description |
|-----------|-------------|
|-----------|-------------|

|      |                                                                                         |
|------|-----------------------------------------------------------------------------------------|
| 9628 | Post-tensioning valve for 700 Bar (max.) single-acting/Power Wedge seater. Wt., 2,4 kg. |
|------|-----------------------------------------------------------------------------------------|

## "TWIN" 4-WAY/3-POSITION (TANDEM CENTER) MANUAL VALVE



Pump-mounted, 6-position detented 5-way manual dual valve. Rated pressure to valve "A" is 700 Bar and valve "B" is 420 bar. Case pressure is 35 bar max.



**Applications:** Multi-strand, double-acting stressing jacks with an auxiliary seating cylinder.

**Actuation:** Dual lever-operated, detent-positioned.

**Operation:**

With valves "A" and "B" in center position, pump is started, and cable is inserted into stressing tool.

Valve "A" is placed in "Stress" position and cylinder extends to tension cable. Pump pressure controls force exerted by tensioning cylinder in this position. "Stress" port is checked internally, and can only be released by building pressure in the valve "B" return position.

When desired cable tension is achieved, valve "A" is placed in valve "B" position and valve "B" in "Seat" position. Seating portion of cylinder will be pressurized to seating pressure controlled by "Seat" relief valve (factory set to 275 bar).

Valve "B" is shifted to "Return" position, which is pressure controlled and will not exceed 155 bar. "Return" portion of stressing tool should be pressurized and will release "Stress" port when pressure reaches 15% of "Stress" port pressure.

"Stress" port will remain open and cylinder will return as long as pressure differential is maintained. "Stress" and "Seat" ports are open to reservoir.

When cylinder has fully returned, both valves are shifted to "Center" position and oil will be directed to reservoir. Maximum pressure setting for the "Seat" relief valve is 420 bar.

**Used on these pumps:** PA17\*, PA46\*, PA55, PE17\*, PE21\*, PE30, PE46\*, PE55, PE120, PE400, PG30\*, PG55, PG120, PG400, PQ60 and PQ120 series.\*

\* These pumps may have reduced first flow stage characteristics due to internal valve restrictions.

| Order No. | Description |
|-----------|-------------|
|-----------|-------------|

|      |                                                                              |
|------|------------------------------------------------------------------------------|
| 9632 | Post-tensioning valve for 700 Bar (max.) double-acting systems. Wt., 6,2 kg. |
|------|------------------------------------------------------------------------------|

**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

**IMPORTANT:** Conversion kit 251528 must be used when mounting any of the valves on this page on PA17 or PE17 pumps.

**IMPORTANT:** When ordering any valve for a PE30 or PG30 series pump, 12,7 mm longer mounting screws are required. For valves 9569, 9570 and 9579, order four 10856 cap screws. For valves 9552, 9572 and 9592, order four 12001 cap screws.



## OIL COOLER KITS

|  | Order No. | Description                                                                          |
|-----------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------|
|                                                                                   | 252511    | Oil cooler kit designed for use with PE604T or PE604PT pumps with 115V. Wt., 2,3 kg. |
|                                                                                   | 252512    | Oil cooler kit designed for use with PE604T or PE604PT pumps with 220V. Wt., 2,3 kg. |

## FLUID LEVEL/TEMPERATURE GAUGE

|  | Description                                                         |                                |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------|
|                                                                                   | Displays fluid level and temperature of hydraulic oil in reservoir. |                                |
|                                                                                   | 0° - 100° C.<br>32 mm wide and 162 mm high.                         |                                |
|                                                                                   | Order No.                                                           | Description                    |
|                                                                                   | 350431                                                              | Fluid level/temperature gauge. |

## RESERVOIR BREATHER KITS

|  | Order No. | Description                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                   | 206767    | Reservoir breather kit designed for use on PA17, PA55, PE17, PE55, PE120, PG55, PG120, PQ60 and PQ120 series pumps. Wt., 0,6 kg.                                                       |
|                                                                                   | 250175    | Reservoir breather kit designed for use on PE21 and PE46 series pumps. These kits replace the reservoir filler cap when the pump is used in dusty and dirty environments. Wt., 0,6 kg. |

## CASTERS

|  | Description                                                                       |                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------|
|                                                                                   | 50,8 mm diameter casters attach to the bottom of large reservoir for portability. |                                   |
|                                                                                   | Sold individually. Order the amount required.                                     |                                   |
|                                                                                   | Order No.                                                                         | Description                       |
|                                                                                   | 10494                                                                             | Single caster wheel. Wt., 0,1 kg. |

## FOOT CONTROL GUARD

|  | Order No. | Description                                             |
|------------------------------------------------------------------------------------|-----------|---------------------------------------------------------|
|                                                                                    | 16339     | Foot Guard for use with 10461 and 251660 foot controls. |

## MAGNETIC STRIP

|  | Order No. | Description                                                                                                                                         |
|-------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                     | 207762    | Magnetic strip with adhesive back can be added to No. 25017, 202777, 202778 and 304718 hand controls. Provides 2,7 kg. of holding force. Wt., 50 g. |

## VITON™ \* SEAL KITS

|  | Description                                                                                                                                                                                                                                                                |                     |       |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------|
|                                                                                     | Viton™* seal kits Can be used in all "C" and "RH" series cylinders, as well as the P12, P55, P59, P157/ P159, P157D/P159D and P300/P300D series of hand pumps. These seals are required when fire resistant hydraulic fluids are used. Not required with Flame-Out® fluid. |                     |       |
|                                                                                     | Order No.                                                                                                                                                                                                                                                                  | Use With            | Model |
|                                                                                     | 300507                                                                                                                                                                                                                                                                     | P12                 | All   |
|                                                                                     | 300472                                                                                                                                                                                                                                                                     | P23, P55            | All   |
|                                                                                     | 300510                                                                                                                                                                                                                                                                     | P59                 | All   |
|                                                                                     | 300508                                                                                                                                                                                                                                                                     | P157, P159, P300    | A     |
|                                                                                     | 300690                                                                                                                                                                                                                                                                     | P157, P159          | B     |
|                                                                                     | 300696                                                                                                                                                                                                                                                                     | P300                | B     |
|                                                                                     | 300508                                                                                                                                                                                                                                                                     | P157D, P159D, P300D | A     |
|                                                                                     | 300693                                                                                                                                                                                                                                                                     | P157D, P159D        | B     |
|                                                                                     | 300699                                                                                                                                                                                                                                                                     | P300D               | B     |

\* Viton™ is a trademark or registered trademark of The Chemours Company

## UNIVERSAL PUMP CART



Mobilize your hydraulic pumps with the PC200. The rugged tubular frame can easily handle pumps weighing up to 91 kg. With 305 mm wheels, the cart rolls easily. Just load the pump onto the cart and wheel it right to the job. The universal mount-hole pattern lets you handle a wide variety of Power Team pumps.

**Cart can be used with the following pumps:** PA60, PA64 and PA554 air/hydraulic pumps; PE55 series, PE183-2 and PE184-2 electric/hydraulic pumps; PE21, PQ60 and PQ120 series "Quiet" pumps; PG55 series gas engine/hydraulic pumps; and pumps with optional 19 L- and 38 L reservoirs; Nos. RP50, RP51, RP101 and RP103.

*(Shown with pump, pump not included)*

| Order No. | Description                                           |
|-----------|-------------------------------------------------------|
| PC200     | Universal pump cart with 305 mm wheels. Wt., 12,2 kg. |

## PROTECTIVE PUMP ROLL CAGE

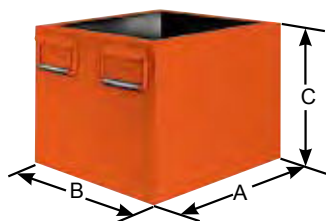


Safeguards pump, gas engine and valves on the job site. Horizontal bars provide convenient hand holds for carrying pump, a pick-up point permits lifting unit with an overhead crane or other device. Standard equipment on PG1203 and PG1204. Can be ordered as an option with any other gas, air, or electrically driven hydraulic pump equipped with a 19 liter reservoir.

**NOTE:** Refer to PG1203/PG1204 pump for details.

| Order No. | Description                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------|
| PC200RC   | Roll cage for use with PC200. (Cannot be used on pumps with 38 liter reservoirs.) Wt., 16,3 kg. |
| RC2GAL    | Roll cage for use with PA46, PA55, PE46, PE55 pumps with 9 liter reservoirs.                    |
| RC5       | Roll cage for PG55 & PG120. For use with PG120 and PG 55 series pumps. Wt., 8,8 kg.             |

## Large capacity reservoirs



Reservoirs are equipped with drain plugs and all necessary conversion items.

Hydraulic oil is not included with reservoir kits. Please order separately.

| Order No. | Cap. (l) | Usable Oil (l/min) | Use With                                         | Size (mm.) |     |     |
|-----------|----------|--------------------|--------------------------------------------------|------------|-----|-----|
|           |          |                    |                                                  | A          | B   | C   |
| RP20**    | 7,6      | 7,1                | PA6, PA50 series (models A-E)                    | 292        | 241 | 165 |
| RP20-F**  | 7,6      | 7,1                | PA6 series (model F), PA 50 series (model F & G) | 292        | 241 | 165 |
| RP20M*    | 9,5      | 7,2                | PA6, PA50 series (models A-E)                    | 292        | 241 | 165 |
| RP20M-F*  | 9,5      | 7,2                | PA6 series (model F), PA50 series (model F & G)  | 292        | 241 | 165 |
| RP21*     | 9,5      | 7,2                | PE18 series                                      | 292        | 241 | 165 |
| RP22†     | 9,5      | 7,1                | PE55, PE120, PA55                                | 292        | 241 | 165 |
| RP50      | 19,0     | 18,4               | PE55, PE120, PA55                                | 457        | 317 | 216 |
| RP51      | 19,0     | 18,4               | PA46, PE21                                       | 457        | 317 | 216 |
| RP100     | 37,9     | 35,1               | PE55, PE120, PA55                                | 457        | 317 | 368 |
| RP101     | 37,9     | 35,1               | PG55, PG120                                      | 457        | 317 | 368 |
| RP103*    | 37,9     | 37,0               | PQ60, PQ120                                      | 392        | 362 | 313 |
| RP104     | 37,9     | 35,1               | PA46, PE46, PE21                                 | 457        | 317 | 356 |

\* Four mounting holes: 1/2"-20, for 50,4 mm diameter swivel casters (No. 10494)

\*\* High density polyethylene reservoir.

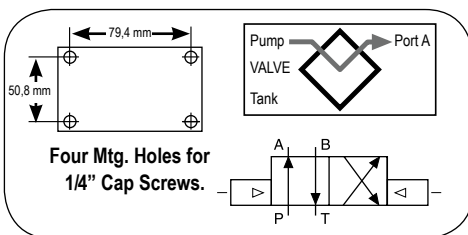
† Aluminum reservoir.

## METAL RESERVOIR CONVERSION KITS FOR PUMPS (INCLUDES GASKETS AND FASTENERS)

| Pump Number | Metal Reservoir Order Number | Metal Reservoir Capacity (l) | Reservoir Wt. (kg) |
|-------------|------------------------------|------------------------------|--------------------|
| PA6         | 213896                       | 1,7                          | 1,4                |
| PA6A        | 213896                       | 1,7                          | 1,4                |
| PA6D        | 213896                       | 1,7                          | 1,4                |
| PA6-2       | 213895                       | 9,5                          | 4,1                |
| PA6D2       | 213895                       | 9,5                          | 4,1                |

| Pump Number | Metal Reservoir Order Number | Metal Reservoir Capacity (l) | Reservoir Wt. (kg) |
|-------------|------------------------------|------------------------------|--------------------|
| PA50        | 213896                       | 1,7                          | 1,4                |
| PA50R       | 213896                       | 1,7                          | 1,4                |
| PA6R        | 213896                       | 1,7                          | 1,4                |
| PA50R2      | 213895                       | 9,5                          | 4,1                |
| PA172       | 213895                       | 9,5                          | 4,1                |

| Pump Number | Metal Reservoir Order Number | Metal Reservoir Capacity (l) | Reservoir Wt. (kg) |
|-------------|------------------------------|------------------------------|--------------------|
| PA174       | 213895                       | 9,5                          | 4,1                |
| PE172       | 213895                       | 9,5                          | 4,1                |
| PE172A      | 213895                       | 9,5                          | 4,1                |
| PE172S      | 213895                       | 9,5                          | 4,1                |
| PE174       | 213895                       | 9,5                          | 4,1                |

Model Shown:  
**9593****3/4-WAY / 2-POSITION SOLENOID ACTUATED VALVES****Application:** Single or double-acting cylinders.**Actuation:** 9593, 9524 and 9554 are solenoid-operated

**Operation with single-acting cylinder:** Either oil port "A" or "B" must be plugged on valve. With port "B" plugged, solenoid is energized to position "A," oil port "A" becomes pressurized. When solenoid is energized to position "B," oil port "A" becomes the return port.

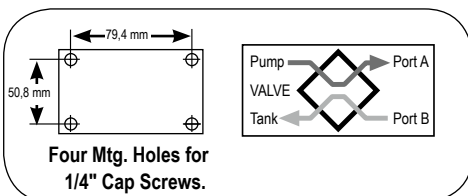
**Operation with multiple single-acting cylinders:** A pressure line from one bank can be connected to oil port "A" and the other to oil port "B" on the valve. Sequence: When energized to position "A," oil port "A" becomes pressurized and clamps the fixture connected to oil port "A". Oil port "B" becomes a "return" port for cylinder connected to oil port "B," and retracts it. The opposite happens when solenoid "B" is energized.

**Operation with double-acting cylinder:** Port "A" is connected to "advance" port of cylinder, oil port "B" connects to cylinder "return" port. Solenoid is energized to position "A," oil port "A" becomes pressurized to extend cylinder piston. The opposite happens when solenoid "B" is energized. Valve does not hold in "retract" position.

**NOTE:** When using more than one valve on a pump, the tank port may require a check valve to permit inadvertent, momentary extension of a retracted cylinder.

**NOTE:** If pump is equipped with an internal outlet check, a "hold" position can be maintained with the pump shut-off.

| Order No. | Description                                                                           |
|-----------|---------------------------------------------------------------------------------------|
| 9593      | 3/4-way 2-position, remote-mounted solenoid valve, 115V, 50/60 Hz., Prod. Wt. 7,0 kg. |
| 9524      | 3/4-way 2-position, remote-mounted solenoid valve, 230V, 50/60 Hz., Wt. 7,0 kg.       |
| 9554      | 3/4-way 2-position, remote-mounted solenoid valve, 24V, 50/60 Hz., Wt. 7,0 kg.        |

Model Shown:  
**9595****3/4-WAY / 2-POSITION AIR ACTUATED VALVES****Application:** Single or double-acting cylinders.**Actuation:** 9595 is air-operated.

**Operation with single-acting cylinder:** Either oil port "A" or "B" must be plugged on valve. With port "B" plugged, solenoid is energized to position "A," oil port "A" becomes pressurized. When solenoid is energized to position "B," oil port "A" becomes the return port.

**Operation with multiple single-acting cylinders:** A pressure line from one bank can be connected to oil port "A" and the other to oil port "B" on the valve. Sequence: When energized to position "A," oil port "A" becomes pressurized and clamps the fixture connected to oil port "A". Oil port "B" becomes a "return" port for cylinder connected to oil port "B," and retracts it. The opposite happens when solenoid "B" is energized.

**Operation with double-acting cylinder:** Port "A" is connected to "advance" port of cylinder, oil port "B" connects to cylinder "return" port. Solenoid is energized to position "A," oil port "A" becomes pressurized to extend cylinder piston. The opposite happens when solenoid "B" is energized. Valve does not hold in "retract" position.

**NOTE:** When using more than one valve on a pump, the tank port may require a check valve to permit inadvertent, momentary extension of a retracted cylinder.

**NOTE:** If pump is equipped with an internal outlet check, a "hold" position can be maintained with the pump shut-off.

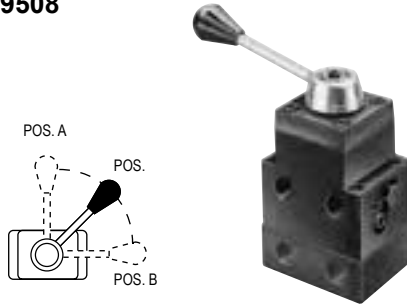
| Order No. | Description                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------|
| 9595      | 3/4-way 2-position, remote-mounted air-operated solenoid valve (minimum of 3,5 bar air pressure required), Prod. Wt. 5,2 kg. |

**CAUTION:** To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.

**CAUTION:** The Posi-Check® feature will not hold the load when shifted directly A to B-B to A or from hold to A or B.

**NOTE:** Maximum tank line pressure for remote-mounted valves is 35 bar.

Model Shown:  
**9508**



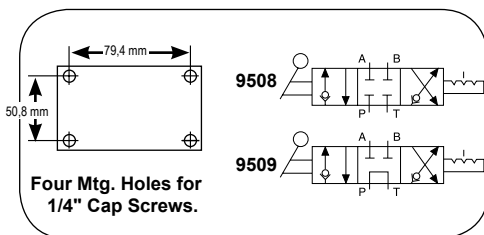
### 4-WAY / 3-POSITION (CLOSED CENTER) AND (TANDEM CENTER) MANUAL VALVES WITH POSI-CHECK®

**Application:** Single or double-acting cylinders. When used with single-acting cylinders, one port must be plugged. For double-acting cylinders, either port can be used to "advance" or "return."

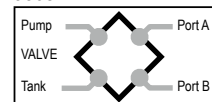
**Actuation:** Lever-operated, detent-positioned.

**Functions:** The 9508 provides "advance," "hold" and "return" positions with all ports blocked (closed center) in the "hold" position. The 9509 has "advance," "hold" and "return" with tandem center (cylinder ports are blocked, pump remains running). Both valves have Posi-Check® feature to guard against pressure loss when shifting from "advance" to "hold."

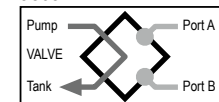
| Order No. | Description                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------|
| 9508      | 4-way 3-position (closed center) manual valve, including sub-plate for remote mounting. Wt., 2,9 kg. |
| 9509      | 4-way 3-position (tandem center) manual valve, including sub-plate for remote mounting. Wt., 2,9 kg. |



**9508**



**9509**



Model Shown:  
**9526**



### 3-WAY / 2-POSITION SOLENOID VALVE

**Application:** Single-acting cylinders.

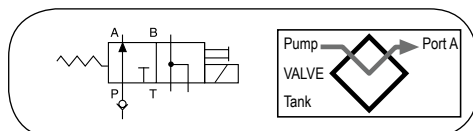
**Actuation:** Solenoid-operated, 115/230/24V, 50/60 Hz.

**Function:** Advances cylinder piston when solenoid is de-energized, and pump is running. When solenoid is energized, oil is directed back through valve "return" port and cylinder piston returns. To place cylinder in "hold" position, pump must be stopped or its flow held at the valve "pressure" port with the solenoid de-energized.

**NOTE:** Valve is equipped with a 9631 snubber valve in port "A." The line from the "return" port of the valve must be unrestricted (100 bar back pressure maximum) back to the reservoir.

**IMPORTANT:** A 9580 in-line check valve must be installed in the "pressure" port if the supply pump is not equipped with an outlet check valve.

**NOTE:** Valves above are shipped without control switch. Use 202777 remote hand switch.



| Order No. | Description                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------|
| 9559      | 3-way 2-position solenoid valve, 115V 50/60 Hz. Includes a remote mounting sub-plate. Wt., 4,4 kg. |
| 9526      | 3-way 2-position solenoid valve, 230V 50/60 Hz. Includes a remote mounting sub-plate. Wt., 4,4 kg. |
| 9556      | 3-way 2-position solenoid valve, 24V 50/60 Hz. Includes a remote mounting sub-plate. Wt., 4,4 kg.  |

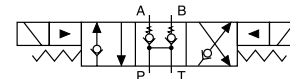
Model Shown:  
**9514**



### 4-WAY / 3-POSITION (TANDEM CENTER) SOLENOID VALVE WITH POSI-CHECK®

**Application:** Double-acting cylinders.

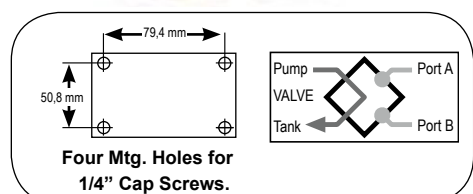
**Actuation:** Solenoid-operated, 115/230/24V, 50/60 Hz.



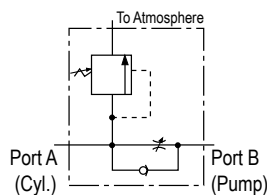
**Functions:** Push button control of "advance," "hold" and "return." The Posi-Check® feature guards against pressure loss when shifting from "advance" to "hold." With valve in "hold" position, cylinder ports are blocked and oil is directed from pump to reservoir.

**NOTE:** Do not allow return tank pressure to exceed 500 bar at the valve, Consult factory before installing a pressure switch on any of these valves.

| Order No. | Description                                                                                                 |
|-----------|-------------------------------------------------------------------------------------------------------------|
| 9514      | 4-way 3-position (tandem center) solenoid valve, 115V, 50/60 Hz. Remote hand control included. Wt., 4,6 kg. |
| 9525      | 4-way 3-position (tandem center) solenoid valve, 230V, 50/60 Hz. Remote hand control included. Wt., 4,6 kg. |
| 9555      | 4-way 3-position (tandem center) solenoid valve, 24V, 50/60 Hz. Remote hand control included. Wt., 4,6 kg.  |



## LOAD-LOWERING VALVE



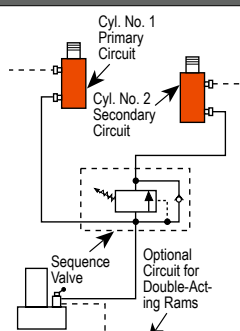
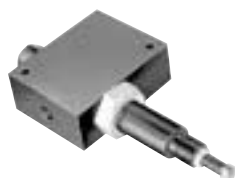
**Application:** Precision metering for controlled cylinder piston return.

**Operation:** Permits free flow when extending cylinder, built-in pressure relief and Posi-Check® locks and holds load in raised position until operator opens valve. May be pre-set to provide consistent metered return, or operator may select rate of return with each actuation. Has 3/8" NPTF ports.

**NOTE:** Pressure relief valve setting is 830 bar. Operating pressure is 700 Bar and max. flow rate is 19 l/min.

| Order No. | Description                     |
|-----------|---------------------------------|
| 9596      | Load-lowering valve. Wt., 1 kg. |

## SEQUENCE VALVE

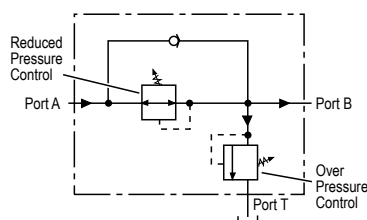


**Application:** Used when one cylinder in a multi-cylinder application must advance before any other.

**Operation:** Pump is connected to port "P" and separate cylinders to ports "A" and "B." When pressure is applied to port "P," cylinder "A" advances. Cylinder "B" will not advance until a predetermined pressure setting is reached in cylinder "A." Pressure setting is adjustable from 35 to 550 bar with adjustment screw (factory preset at 70 bar). Has 3/8" NPTF ports.

| Order No. | Description                                     |
|-----------|-------------------------------------------------|
| 9597      | Pressure control sequencing valve. Wt., 2,5 kg. |

## PRESSURE REDUCING VALVE



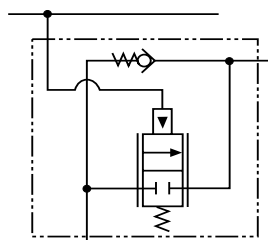
**Application:** Provides complete, independent pressure control to two or more clamping systems operated by a single power source.

**Operation:** Can be used to provide different pressures in various stages of a single system. Virtually zero leakage across valve means each system can be operated by a single continuous pressure source. Adjustable from 70 to 350 bar at outlet port "B" (secondary). Has 1/4" NPTF ports.

| Order No. | Description                           |
|-----------|---------------------------------------|
| 9608      | Pressure reducing valve. Wt., 2,6 kg. |

**CAUTION:** Over Pressure control must be set at a higher value than operating pressure.

## COUNTER BALANCE VALVE



**Application:** Double-acting cylinders. Provides positive holding and controlled, "chatter-free" lowering of a load.

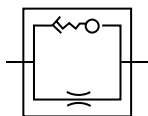
**Operation:** Load is raised at flow rate of pump, and held when pump is shut-off. When the pump is shifted to "retract," the counter balance valve will continue to hold the load until system pressure exceeds pressure caused by load. The load can then be lowered smoothly to the flow rate of the pump. The counter balance valve is designed to operate with pumps having a high pressure flow rate of up to 1,9 l/min. and cylinder ratios of 3 to 1.

| Order No. | Description                                                                                                                                          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9720      | Counter balance valve, including two pairs of fittings (male and female), hydraulic hoses, and dust caps. Wt., 4,5 kg.                               |
| 9721      | Counter balance valve, including two pairs of fittings (male and female), but does not include couplers, hoses, fittings and dust caps. Wt., 4,2 kg. |

**CAUTION:** The 9720 patented counter balance valve has a pilot pressure as high as 210 bar. Because this pressure is applied to the rod end of the cylinder while it is already under load, the system should not be sized for loads greater than 80% of cylinder rated capacity. To prevent sudden, uncontrolled descent of a load as it is being lowered, use a 9596 Load-Lowering Valve or 9720 Counter Balance Valve in conjunction with the directional valve used in your application.



## METERING VALVE



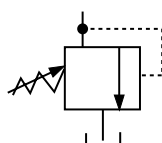
**Application:** For systems using large cylinders or extended lengths of hydraulic hose.

**Operation:** Controls surges by restricting flow if it exceeds 26,5 l/min. When flow subsides, valve reopens automatically. Has 3/8" NPTF male end to thread into return port of system control valve, and a 3/8" NPTF female end, permitting return hose to be directly connected.

| Order No. | Description |
|-----------|-------------|
|-----------|-------------|

|      |                              |
|------|------------------------------|
| 9631 | Metering valve. Wt., 0,1 kg. |
|------|------------------------------|

## "IN-LINE" PRESSURE REGULATOR VALVE



**Application:** Single or double-acting cylinders. Permits adjusting operating pressures at various values below relief valve setting of pump.

**Operation:** Regulator valve is easily adjusted to maintain pressures between 20 and 700 Bar. Maintains a given pressure setting within 3% over repeated cycles. Flow range is 0,3 l/min to 23 l/min.

| Order No. | Description |
|-----------|-------------|
|-----------|-------------|

|      |                                                                                                                              |
|------|------------------------------------------------------------------------------------------------------------------------------|
| 9633 | In-line pressure regulator valve with two 3/8" NPTF inlet ports, one 1/8" NPTF tank port and 1 m drain line kit. Wt., 1.9 kg |
|------|------------------------------------------------------------------------------------------------------------------------------|

## RELIEF VALVE



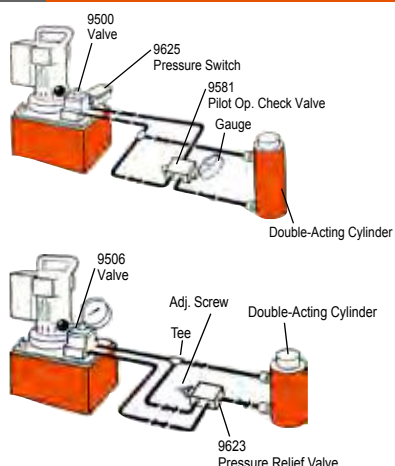
**Application:** Provide an economical means of protecting a hydraulic circuit against over pressurization.

**Operation:** These factory preset valves are designed for maximum flow rate of 19 l/min. Furnished with 1/8" NPTF male port. All valves weigh 0,1 kg. See chart below for ordering information.

**NOTE:** Care should be exercised to protect workers from hot, pressurized hydraulic oil. Install these valves only in an enclosed or shielded area.



## Typical Set-Up



| Relief Valve Order No. | Pressure Setting* (bar) | Relief Valve Order No.                 | Pressure Setting* (bar) |
|------------------------|-------------------------|----------------------------------------|-------------------------|
| RV21278                | 697/738                 | RV21278-50                             | 352/393                 |
| RV21278-6              | 41/44                   | RV21278-52                             | 366/407                 |
| RV21278-10             | 62/69                   | RV21278-55                             | 386/428                 |
| RV21278-15             | 103/117                 | RV21278-57                             | 400/442                 |
| RV21278-17             | 110/124                 | RV21278-60                             | 421/462                 |
| RV21278-20             | 131/152                 | RV21278-65                             | 455/497                 |
| RV21278-25             | 159/186                 | RV21278-70                             | 490/531                 |
| RV21278-27             | 179/193                 | RV21278-75                             | 524/566                 |
| RV21278-28             | 186/207                 | RV21278-80                             | 559/600                 |
| RV21278-30             | 207/235                 | RV21278-83                             | 580/621                 |
| RV21278-32             | 214/228                 | RV21278-86                             | 600/642                 |
| RV21278-35             | 241/262                 | RV21278-88                             | 614/662                 |
| RV21278-38             | 283/310                 | RV21278-90                             | 628/669                 |
| RV21278-40             | 304/331                 | RV21278-114                            | 793/834                 |
| RV21278-43             | 338/366                 | RV21278-6280                           | 440/476                 |
| RV21278-48             | 352/393                 | *All RV's are Preset - Non-Serviceable |                         |



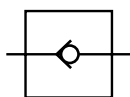
## SHUT-OFF VALVE



**Application:** This needle valve permits fine metering of hydraulic oil  
**Operation:** Can be used for controlling multiple single-acting cylinders.

| Order No. | Description                                       |
|-----------|---------------------------------------------------|
| 9575      | Shut off valve with 3/8" NPTF ports. Wt., 0,6 kg. |

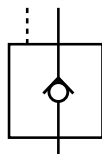
## CHECK VALVE



**Application:** Permits flow of hydraulic oil in one direction only.  
**Operation:** Installs right in hydraulic line.

| Order No. | Description                                        |
|-----------|----------------------------------------------------|
| 9580      | Check valve with 3/8" NPTF male ends. Wt., 0,2 kg. |

## PILOT-OPERATED CHECK VALVE

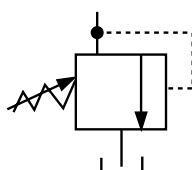


**Application:** For use with open or tandem center valves. Permits free flow of fluid in one direction.

**Operation:** Flow is blocked in opposite direction until pilot oil pressure is applied. This prevents the loss of pressure if the valve is inadvertently shifted or the pump line is broken. Minimum cracking pressure is 4,1 bar. Required pilot pressure is approximately 16% of checked system pressure.

| Order No. | Description                                                   |
|-----------|---------------------------------------------------------------|
| 9581      | Pilot-operated check valve with 3/8" NPTF ports. Wt., 1,7 kg. |

## "IN-LINE" PRESSURE RELIEF VALVE



**Application:** Single or double-acting cylinders. For remote locations in a hydraulic circuit where maximum pressure requirements are less than basic overload valve setting in pump.

**Operation:** Adjustable from 70 to 700 Bar. Valve is spring-loaded and direct-acting.

| Order No. | Description                                            |
|-----------|--------------------------------------------------------|
| 9623      | Pressure relief valve with 3/8"NPTF ports. Wt. 0,9 kg. |

Model Shown:  
9639, 9638



### Performance Requirements

For additional technical information or to order a **Material Safety Data Sheet** call 1-800-477-8326 or go to [www.powerteam.com](http://www.powerteam.com).

### Features

#### STANDARD HYDRAULIC OIL

- For dependable performance of all your hydraulic pumps and cylinders.
- Contains foam suppressant additives and has a high viscosity index.

#### FLAME-OUT® 220 FIRE RESISTANT HYDRAULIC FLUID\*

- Contains anti-rust, anti-foam and anti-sludge additives.
- Provides fire resistant protection. Note: Will burn if heat source is extreme enough. Will not, however, propagate the flame and is self-extinguishing when there is no ignition source.
- Provides maximum lubrication and heat transfer.
- Offers a wider operating temperature range.
- No need to change seals in your Power Team equipment. Just drain the standard oil and replace it with Flame-Out® 220.

#### BIODEGRADABLE HYDRAULIC FLUID

- Biodegradable, non-toxic fluid withstands moderate to severe operating conditions. Provides excellent protection against rust.
- Offers superior anti-wear properties and has excellent multi-metal compatibility.

#### LOW TEMPERATURE OIL

- Provides smooth, reliable operation in the coldest climate conditions.

### Technical Specifications

| Description   | Grade<br>(ASTM) | Specific Gravity<br>@ 16°C<br>(kg/l) | Color<br>(ASTM) | Flash Point | Fire Point | Pour Point | Viscosity  |            | Viscosity Index | Foam Test<br>(ASTM) |
|---------------|-----------------|--------------------------------------|-----------------|-------------|------------|------------|------------|------------|-----------------|---------------------|
|               |                 |                                      |                 |             |            |            | SUS @ 38°C | SUS @ 99°C |                 |                     |
| Standard Oil  | 215             | 0,88                                 | 2 Amber         | 204°C       | 221°C      | -34°C      | 215        | 48         | 100 min.        | Pass                |
| Flame-Out®    | 220             | 0,91                                 | Light Amber     | 260°C       | 288°C      | -26°C      | 220        | 55         | 140 min.        | Pass                |
| Biodegradable | —               | 0,92                                 | 2 Amber         | 224°C       | NA*        | -30°C      | 183        | 53         | 213 min.        | Pass                |
| Low Temp.     | —               | 0,87                                 | 6.5 Red         | 180°C       | 204°C      | -45°C      | 183        | 52         | 190 min.        | Pass                |

### Ordering Information

| Order Number | Oil Description | Quantity l |
|--------------|-----------------|------------|
| 9636         | Standard Oil    | 0,9        |
| 9637         |                 | 3,8        |
| 9638         |                 | 9,5        |
| 9616         |                 | 208        |
| 9639         | Flame-Out®      | 3,8        |
| 9640         |                 | 9,5        |
| 9645         | Biodegradable   | 3,8        |
| 9646         |                 | 9,5        |
| 9647         | Low Temperature | 3,8        |

## ON/OFF MOTOR CONTROL



25017



203225



10461



251660

The following remote control switches will give you momentary "ON" control of your hydraulic pump. These switches are deadman type, spring loaded to the "OFF" position. They can be used with any Power Team electric hydraulic pump.

| Order No. | Description                                                                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25017     | Remote hand control. Has a push button switch, with a 3 m cord. Wt., 0,4 kg.                                                                                                         |
| 203225    | Remote hand control. Heavy-duty with single push button switch in a neoprene housing with 3 m cord. Housing seals out dust, lint and liquids (unit is not submersible). Wt., 0,4 kg. |
| 10461     | Remote foot control, with 3 m cord. Wt., 1,4 kg.                                                                                                                                     |
| 251660    | Remote foot control, with 3 m cord. For use with the PE10 style pumps. Wt., 0,5 kg.                                                                                                  |

## SOLENOID &amp; MOTOR CONTROL

202777  
202778  
304718309652  
309653

209593

17627  
216209

| Order No. | Description                                                                                                                                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 202777 *  | Remote hand control. Has rocker style switch that is momentary advance, spring center hold and detented retract. It comes with a 3 m cord, for use with 3-way/2 or 3-position valves. Wt., 0,4 kg.                                                                                |
| 202778 ** | Remote hand control. Has rocker style switch that is momentary advance, spring center hold and momentary retract. It comes with a 3 m cord, for use with 4-way/3-position valves. Wt., 0,4 kg.                                                                                    |
| 304718**  | Remote hand control. Has a rocker style switch that is momentary advance, spring center hold and momentary retract. The switch is wired to start and stop the motor when the valve is energized. It comes with a 3 m cord. To be used with 4-way/2-position valves. Wt., 0,4 kg.  |
| 309653    | Remote foot control. Can be used in place of either of the above hand controls to control the same type of valves. The switch is momentary on, both advance and retract position, and is spring centered to the hold position. This foot switch comes with 3 m cord. Wt., 1,8 kg. |
| 17627     | Remote foot control. Same as the 309653, but without a cord. Wt., 0,9 kg.                                                                                                                                                                                                         |
| 309652    | Remote foot control. Has same functions as 304718. Supplied with a 3 m cord. To be used with 4-way/2-position valves. Wt., 1,8 kg.                                                                                                                                                |
| 216209    | Remote foot control. Same as the 309652, but without a cord. Wt., 0,9 kg.                                                                                                                                                                                                         |

\* For use on solenoid valves that are used on single-acting cylinders

\*\* For use on solenoid valves that are used on double-acting cylinders

NOTE: See valves listing to determine which remote to use.

## REMOTE AIR MOTOR CONTROLS



209593

This remote hand control has two momentary push buttons, one for advance and one for retract with spring offset to hold. To be used with 4-way/2-position air pilot valves.

| Order No. | Description                                       |
|-----------|---------------------------------------------------|
| 209593    | Remote hand control with 3,7 m cord. Wt., 0,9 kg. |

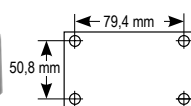
## SUB-PLATES



9510



9620



9510 and 9620 attach to the bottom of valve for remote mounting.

For remote mounting of control valves. Sub-plates convert pump-mounted valves to remote-mounted valves quickly and easily.

| Order No. | Description                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 9510      | Sub-plate for remote mounting the following valves; 9500, 9501, 9502, 9504, 9506, 9507, 9511, 9552, 9572, 9575, 9576, 9592 and 9594. Wt., 0,7 kg. |
| 9620      | For use with 9500, 9501, 9502, 9552, 9572, 9592 and 9594. Same as 9510, but has integral pressure regulating valve. Wt., 1,7 kg.                  |

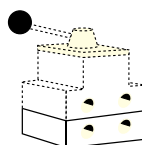
## PUMP-MOUNTED SUB-PLATES



9515



9521



The 9515 and 9521 mount between the pump cover plate and valve.

When fitted between pump cover plate, valve mounting flange, and control valve, provides a separate 3/8" NPTF female port, open to "return" regardless of position of valve. Also provides a separate 3/8" NPTF female pressure port. This sub-plate can be useful when you desire to use one pump with a deck-mounted control valve, plus a separate remote-mounted valve to control another function.

**For use with the following valves:** 9500, 9501, 9502, 9504, 9506, 9507, 9511, 9520, 9552, 9572, 9575, 9576, 9592, 9594, and 9609.

| Order No. | Description                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9515      | Sub-plate, Wt., 0,6 kg.                                                                                                                                         |
| 9521      | Sub-plate for use under most pump-mounted valves to provide adjustable pressure control on units not equipped with an external pressure regulator. Wt., 1,7 kg. |

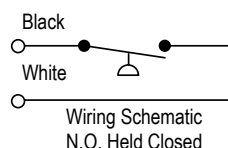
## AIR FILTER/REGULATOR/LUBRICATOR



Recommended for use with single-speed air/hydraulic pumps.

| Order No. | Description                                                |
|-----------|------------------------------------------------------------|
| 9531      | Filter/regulator. 1/4" NPTF inlet and outlet. Wt., 0,4 kg. |

## PRESSURE SWITCH



**Application:** Used in a hydraulic circuit where system pressure must be "held." Automatically (electrically) turns off pump motor when predetermined system pressure is reached.

Attaches directly to control valve manifold or can be mounted in-line to read system pressure. Has a 1/4" NPTF male thread, and a 1/4" NPTF fitting for gauge mounting if required. Adjustable from 70 to 700 Bar. Can also be used to actuate other electrical devices in the system. Wired normally open and held closed by spring pressure.

| Order No. | Description                                                     |
|-----------|-----------------------------------------------------------------|
| 9625      | In-line pressure switch with 1/4" NPTF gauge port. Wt., 0,5 kg. |

**IMPORTANT:** Electrical rating of switch is 5 amps at 250V max. To prevent permanent damage to switch, a control relay must be installed to handle currents or voltage exceeding these limits. Pressure switch should never be used to directly actuate the electrical motor.

## PILOT-OPERATED AIR CONTROL VALVES



**Application:** For use when an air pilot signal is required at a set hydraulic pressure. Can be used to shift valves, and start or stop pneumatic pumps.

Attaches directly to control manifold or can be mounted in-line to read system hydraulic pressure. Automatically turns on an air pilot signal when a predetermined system pressure is reached. Has 1/4" NPTF male thread and 1/4" NPTF fitting for gauge mounting if required. Adjustable from 35 - 700 Bar. Maximum rating of 700 l at 700 bar.

| Order No. | Description                                                                             |
|-----------|-----------------------------------------------------------------------------------------|
| 9641      | Pilot-operated control valve, normally closed, with 1/4" NPTF male thread. Wt., 0,5 kg. |
| 9643      | Pilot-operated control valve, normally open, with 1/4" NPTF male thread. Wt., 0,5 kg.   |

Model Shown:

9755E



Accessories

## Features

## REINFORCED WIRE-BRAID RUBBER HOSES OFFER INCREASED DURABILITY

- Hose reinforced with 4 braids of high tensile steel wire.
- The synthetic rubber covering is oil and weather resistant.
- 3/8" NPTF fittings on both ends.
- Operating pressure is 700 Bar.



## Cylinder Return Time

| Cylinder No. | 9769E                     | 9781E                     |
|--------------|---------------------------|---------------------------|
|              | 3,1 m Hose<br>6,4 mm I.D. | 3,1 m Hose<br>9,5 mm I.D. |
| C2514C       | 51 sec.                   | 14 sec.                   |
| C556C        | 1 min., 30 sec.           | 24 sec.                   |
| C5513C       | 4 min., 12 sec.           | 59 sec.                   |
| C10010C      | 6 min., 56 sec.           | 1 min. 3 sec.             |



## Hose with Coupler Half

9754E Hose assembly consisting of 9756E (1,8 m hose) with 9798 (hose half coupler) and 9800 (dust cap)

## Ordering Information

| Hose Type          | Hose I.D.<br>(mm) | Hose Length<br>(m) | Burst Rating<br>(bar) | Order Number |
|--------------------|-------------------|--------------------|-----------------------|--------------|
| Rubber, Wire-Braid | 6,5               | 0,9                | 3250                  | 9755E        |
| Rubber, Wire-Braid | 6,5               | 1,8                | 3250                  | 9756E        |
| Rubber, Wire-Braid | 6,5               | 1,8                | 3250                  | 9754E**      |
| Rubber, Wire-Braid | 6,5               | 2,4                | 3250                  | 9757E        |
| Rubber, Wire-Braid | 6,5               | 3,1                | 3250                  | 9758E        |
| Rubber, Wire-Braid | 6,5               | 3,7                | 3250                  | 9759E        |
| Rubber, Wire-Braid | 6,5               | 6,1                | 3250                  | 9760E        |
| Rubber, Wire-Braid | 6,5               | 9,1                | 3250                  | 9761E        |
| Rubber, Wire-Braid | 6,5               | 15,3               | 3250                  | 9762E        |

\*\*Furnished with 9798 hose half coupler and 9800 dust cap.

| Hose Type           | Hose I.D.<br>(mm) | Hose Length<br>(m) | Burst Rating<br>(bar) | Order Number |
|---------------------|-------------------|--------------------|-----------------------|--------------|
| Rubber, Wire-Braid* | 9,5               | 0,9                | 2800                  | 9733E        |
| Rubber, Wire-Braid* | 9,5               | 1,8                | 2800                  | 9776E        |
| Rubber, Wire-Braid* | 9,5               | 3,1                | 2800                  | 9777E        |
| Rubber, Wire-Braid* | 9,5               | 4,6                | 2800                  | 9734E        |
| Rubber, Wire-Braid* | 9,5               | 6,1                | 2800                  | 9778E        |
| Rubber, Wire-Braid* | 9,5               | 9,1                | 2800                  | 9735E        |
| Rubber, Wire-Braid* | 9,5               | 12,2               | 2800                  | 9736E        |
| Rubber, Wire-Braid* | 9,5               | 15,3               | 2800                  | 9779E        |

\* High Flow



Model Shown:  
9767E



### Features

#### POLYURETHANE HOSES.

- Nylon core tube with high tensile steel wire reinforcement.
- Hoses are equipped with plastic hose guards.
- Ben radius 70 mm
- Suitable for continuous service at temperatures from -40° to 100° C.



#### Cylinder Return Time

| Cylinder No. | 9769E                     | 9781E                     |
|--------------|---------------------------|---------------------------|
|              | 3,1 m Hose<br>6,4 mm I.D. | 3,1 m Hose<br>9,5 mm I.D. |
| C2514C       | 51 sec.                   | 14 sec.                   |
| C556C        | 1 min., 30 sec.           | 24 sec.                   |
| C5513C       | 4 min., 12 sec.           | 59 sec.                   |
| C10010C      | 6 min., 56 sec.           | 1 min. 3 sec.             |



#### BE SAFE!

1. Inspect before and after each use.
2. Replace if any signs of degradation or wear that may affect safety or performance.
3. Keep the product clean.
4. Store properly.

### Ordering Information

| Hose Type    | Hose I.D.<br>(mm) | Hose Length<br>(m) | Burst Rating<br>(bar) | Order Number |
|--------------|-------------------|--------------------|-----------------------|--------------|
| Polyurethane | 6,4               | 0,6                | 2800                  | 9765E        |
| Polyurethane | 6,4               | 0,9                | 2800                  | 9766E        |
| Polyurethane | 6,4               | 1,8                | 2800                  | 9767E        |
| Polyurethane | 6,4               | 1,8                | 2800                  | 9764E**      |
| Polyurethane | 6,4               | 2,4                | 2800                  | 9768E        |
| Polyurethane | 6,4               | 3,1                | 2800                  | 9769E        |
| Polyurethane | 6,4               | 3,6                | 2800                  | 9770E        |
| Polyurethane | 6,4               | 6,1                | 2800                  | 9771E        |
| Polyurethane | 6,4               | 15,3               | 2800                  | 9772E        |
| Polyurethane | 6,4               | 22,9               | 2800                  | 9750E        |
| Polyurethane | 6,4               | 30,5               | 2800                  | 9751E        |

| Hose Type     | Hose I.D.<br>(mm) | Hose Length<br>(m) | Burst Rating<br>(bar) | Order Number |
|---------------|-------------------|--------------------|-----------------------|--------------|
| Polyurethane* | 9,5               | 1,8                | 2800                  | 9780E        |
| Polyurethane* | 9,5               | 3,1                | 2800                  | 9781E        |
| Polyurethane* | 9,5               | 6,1                | 2800                  | 9782E        |
| Polyurethane* | 9,5               | 15,3               | 2800                  | 9783E        |

\* High Flow

\*\*Furnished with 9798 hose half coupler and 9800 dust cap.

Model Shown:

9775



Accessories

## Features

## NON-CONDUCTIVE HOSE

- For applications requiring electrical isolation.
- 3/8" NPTF fittings on both ends
- Leakage factor of less than 50 micro-ampere.
- Orange polyurethane for easy identification.
- Covering is not perforated, preventing moisture from entering the hose and affecting its overall conductivity.
- Hoses feature a minimum 2800 bar burst pressure.



## BE SAFE!

1. Inspect before and after each use.
2. Replace if any signs of degradation or wear that may affect safety or performance.
3. Keep the product clean.
4. Store properly.



## Torque Wrench Hoses - Non-Conductive Dual Line



| Order Number | Hose Length (m) | Hose I.D. (mm) |
|--------------|-----------------|----------------|
| TWH15        | 4,6             | 6,4            |
| TWH20        | 6               | 6,4            |
| TWH30        | 15,2            | 6,4            |

For torque wrench tools, refer to the tool section.

## Ordering Information

| Order Number | Couplers / Fitting | Hose I.D. (in.) | Hose Length (m) | Burst Rating (bar) |
|--------------|--------------------|-----------------|-----------------|--------------------|
| 9773         | 3/8" Fitting NPTF  | 6,4             | 1,8             | 2800               |
| 9774         | 3/8" Fitting NPTF  | 6,4             | 3,1             | 2800               |
| 9775         | 3/8" Fitting NPTF  | 6,4             | 6,1             | 2800               |

**CYLINDER AND HOSE COUPLERS** are designed for use up to 700 bar with hydraulic jacks, cylinders, etc. They are the threaded union type for interchanging cylinders in seconds. Each half is valved with a precision ball for a tight shutoff when disconnected. These couplers also permit the separation of cylinders or hose from pump when at 0 bar with minimal oil loss.

**NO SPILL, PUSH TO CONNECT COUPLERS** are designed to permit high oil flow, the no-spill, push-to-connect couplers with locking collar and flush face design are for high pressure applications. The flush-face concept makes it easy to clean both coupler ends before connecting. Our unique push-to-connect, "dry-break" design eliminates oil spillage. The locking collar makes accidental disconnects a thing of the past. For 700 bar operation.

|                                                                                     | Order No. | Description                                                                           |                                                                                      | Order No. | Description                                                                                                                              |
|-------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------|
|    | 9795      | Complete quick coupler with two 9800 dust caps, 3/8" NPTF.                            |     | 251410    | Quick-connect, screw-on female coupling. Used on 700 bar torque wrenches, nut-splitters, hoses, and pumps.<br><br>1/4" NPT Female Thread |
|   | 9798      | Male (hose) half coupler 3/8" NPTF. Includes half dust cap.                           |    | 251411    | Quick-connect, screw-on male nipple. Used on 700 bar torque wrenches, nut-splitters, hoses, and pumps.<br><br>1/4" NPT Female Thread     |
|  | 9796      | Female (cylinder) half coupler with 9800 dust cap, 3/8" NPTF.                         |  | 252364    | Metal dust cover for female coupler                                                                                                      |
|  | 9796-V*   | Female (cylinder) half coupler with Viton™ seals and 9797 metal dust cap, 3/8" NPTF.  |  | 252365    | Metal dust cover for male coupler                                                                                                        |
|  | 9796-E    | Female (cylinder) half coupler with EPR seals and No. 9797 metal dust cap, 3/8" NPTF. |  | 9792      | No-spill, push-to-connect, Female (cylinder) half quick coupler only. (Wt., 0,14 kg.)                                                    |
|  | 9799      | Optional metal dust cap (hose half).                                                  |  | 9793      | No-spill, push-to-connect, Male (hose) half quick coupler only. (Wt., 0,14 kg.)                                                          |
|  | 9797      | Optional metal dust cap for cylinder half.                                            |  | 9794      | No-spill, push-to-connect, Complete quick coupler (male and female). Dust caps not included. (Wt., 0,23 kg.)                             |
|  | 9800      | Dust cap for male or female 3/8" NPTF half couplers. (Wt., 0,14 kg.)                  |                                                                                      |           |                                                                                                                                          |

\* Viton™ is a trademark or registered trademark of The Chemours Company

Model Shown:

9040E



### Accessories for Analog Gauges



#### 9046 - Silicone Fill Kit

Silicone fill kit. 212 gr. Requires one bottle to fill 102 mm gauge; four bottles to fill 152 mm gauge.



#### 9049 - Dampener Fitting

High-performance pulsation dampener. 1/4" NPTF male x 1/4" NPTF female.

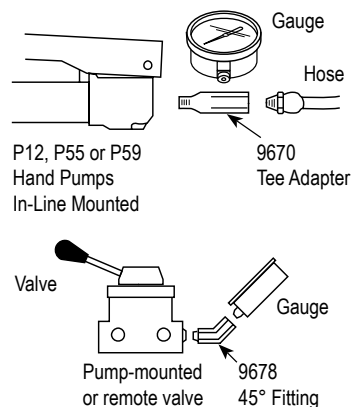
### Features

#### HEAVY-DUTY HYDRAULIC PRESSURE GAUGES

- Gauges feature an easily readable and highly visible, red Day-glo® needle.
- High strength steel bourdon tube ensures high cycle life.
- Stainless steel cases and lens locking rings.
- Have 1/4" NPT connections.
- Gauges are calibrated and comply to ASME B40.1 GRADE B



#### Typical Set-Up - Mounting Situations



### Ordering Information

| Order Number | Face Dia. (mm) | Pressure Range (bar) | Tonnage Range (ton) | Major Graduations (bar)          | Minor Graduations (bar)                                          | Silicone Filled (Yes/No) | Use With Cylinder Series               |
|--------------|----------------|----------------------|---------------------|----------------------------------|------------------------------------------------------------------|--------------------------|----------------------------------------|
| 9040E        | 63,5           | 0 - 690              | —                   | 2000 psi, 100 bar                | 200 psi, 20 bar                                                  | Yes                      | All                                    |
| 9052E        | 101,6          | 0 - 690              | —                   | 1000 psi, 100 bar                | 100 psi, 20 bar                                                  | Yes                      | All                                    |
| 9053E        | 101,6          | 0 - 690              | 0-5                 | 2000 psi, 1 Ton                  | 200 psi, 0,1 Ton                                                 | No                       | C & RLS                                |
| 9055E        | 101,6          | 0 - 690              | 0-10                | 2000 psi, 1 Ton                  | 200 psi, 0,1 Ton                                                 | No*                      | C, RD, RH, RLS & RSS                   |
| 9057E        | 101,6          | 0 - 690              | 0-15                | 2000 psi, 1 Ton                  | 200 psi, 0,2 Ton                                                 | Yes                      | C                                      |
| 9059E        | 101,6          | 0 - 690              | 0-17,5, 0-30, 0-50  | 2000 psi, 5 Ton                  | 200 psi, 0,5 Ton on 30, 50 Ton Scales; 0,2 Ton on 17,5 Ton Scale | No*                      | RT172, RT302, RT503                    |
| 9061E        | 101,6          | 0 - 690              | 0-20                | 2000 psi 5 Ton                   | 200 psi, 0,5 Ton                                                 | Yes                      | RH, RLS, RSS                           |
| 9063E        | 101,6          | 0 - 690              | 0-25                | 2000 psi 5 Ton                   | 200 psi, 0,5 Ton                                                 | No*                      | C & RD                                 |
| 9065E        | 101,6          | 0 - 690              | 0-30                | 2000 psi, 5 Ton                  | 200 psi, 0,5 Ton                                                 | No*                      | RH†, RLS & RSS                         |
| 9067E        | 101,6          | 0 - 690              | 0-50                | 2000 psi, 5 Ton                  | 200 psi, 0,5 Ton                                                 | No*                      | RH†, RLS & RSS                         |
| 9069E        | 101,6          | 0 - 690              | 0-55                | 2000 psi, 5 Ton                  | 200 psi 0,5 Ton                                                  | No*                      | C, R, RA & RD                          |
| 9071E        | 101,6          | 0 - 690              | 0-60                | 2000 psi, 5 Ton                  | 200 psi, 1 Ton                                                   | No*                      | RH, except RH6010                      |
| 9073E        | 101,6          | 0 - 690              | 0-75                | 2000 psi, 5 Ton                  | 200 psi, 1 Ton                                                   | Yes                      | C, RLS & RD8013                        |
| 9075E        | 101,6          | 0 - 690              | 0-100               | 2000 psi, 10 Ton                 | 200 psi, 1 Ton                                                   | No*                      | C, R, RA, RD, RH, RLS†, RSS† & RT1004† |
| 9077E        | 101,6          | 0 - 690              | 0-150               | 2000 psi, Initial 10 Then 20 Ton | 200 psi, 2 Ton                                                   | No*                      | C, R, RD & RLS                         |
| 9079E        | 101,6          | 0 - 690              | 0-200               | 2000 psi, 20 Ton 10 Then 20 Ton  | 200 psi, 2 Ton                                                   | No*                      | R, RD & RH†                            |
| 9089         | 152,4          | 0 - 690              | 0-690               | 1000 psi, 100 bar                | 100 psi, 10 bar                                                  | No                       | All                                    |

\* Shipped "dry." User can convert to "wet" using liquid silicone No. 9046.

† The tonnage scale on the gauge may vary slightly among different series cylinders due to different effective area.

Model Shown:  
**DG100**



### Features

#### DG100 DIGITAL PRESSURE GAUGES

- Accurate to within 1%, Long-life pressure transducer.
- Pressure values are displayed on large red LEDs in 0,7 bar or bar increments.
- "Peak" hold feature with reset toggle switch and "Peak On" indicator; Hi/Low set point feature with relay outputs for Hi/Low alarms and/or control signals.
- A slow flashing display indicates pressure below the low limit. Fast blinking display alerts if limit is exceeded.
- High and low limit relays are rated to 5 amps at 115V.
- Operating temperature of -18 to 60°C for the electronic display and -29 to 82°C for the transducer.
- Gauge housings are extruded aluminum 1/8" DIN enclosures (NEMA 1 rating).
- When power cable is connected to gauge, display will scroll all characters, performing a self- diagnostic routine.
- 1/4" NPTF male threads for the pressure connection.
- 6-foot input signal cable connects to back of display unit.

Accessories



#### Typical Set-Up - Mounting Situations



- **DG100** (bar) digital gauge.
- **DG100B** (bar) digital gauge.

Upgrade to **DG100** or **DG100B** when:

- Hi/Low alarms required.
- Relays triggered by Hi/Low pressure.
- Pressure display needs to be remotely mounted from measurement location.
- Accurate within 1% @ Full Scale.



#### Accessories for DG100



##### 420778 - Digital Gauge Stand

Gauge stand for DG100. Has angled base mounting to hold gauge at a convenient viewing angle. Wt., 0,5 kg.



##### 37045 - Auxiliary Power Cord

Auxiliary power cord for use with any 12 or 24V battery. Wt., 0,1 kg. *Caution: For use on negative ground systems only.*

#### Ordering Information

| Order Number    | Description            | Pressure Range (bar) | Prod. Wt. (kg) |
|-----------------|------------------------|----------------------|----------------|
| <b>DG100 *</b>  | Digital pressure gauge | 0 - 700              | 1              |
| <b>DG100B *</b> | Digital pressure gauge | 0 - 700              | 1              |

\* Serviced only at factory.



Model Shown:

9042DG, 9042DG1500



Accessories

## Features

- Digital gauge is easier to read and offers better accuracy than a conventional analog gauge.
- The laser welded stainless steel sensor & socket and the IP67 weatherproof rating make this product suitable for use in the most demanding of applications.
- Five pre-programmed engineering units allow technicians to read pressure in the unit of measure most applicable to the process.
- The gauge also features a bar graph display feature for enhanced visibility.
- Includes automatic-off battery conservation, pressure tare, minimum pressure memory and maximum pressure.
- Vibration & shock tested to MIL-STD-202G.
- Agency Compliance / Approval: RoHS, CE, ASME B40.7, UL, cUL 61010-1.
- The gauges are calibrated for life at the factory and can be certified in the field, if required.

Power Team's 9042DG was the perfect choice in accurately monitoring the force applied in this H frame press application.

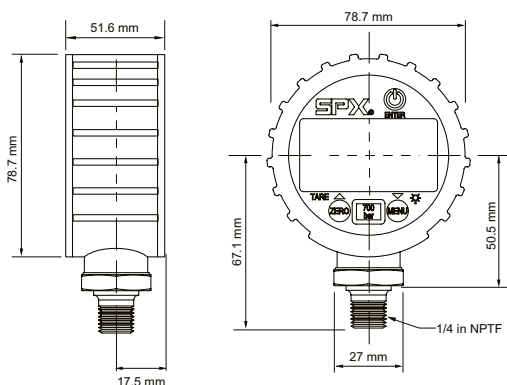


### Technical Attributes



- |                                                                                   |                                                                                              |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1 Protective rubber boot                                                          | 5 2,000 hr. life, (2) AA (LR6) batteries                                                     |
| 2 Back-light and large 12,2 mm display easy reading                               | 6 Typical Cycle Life: 10,000,000                                                             |
| 3 Displays in multiple engineering units: psi, bar, mPa, inHg, kg/cm <sup>2</sup> | 7 1/4" NPTF Male Threads (9042DG); 9/16-18 UNF-2B (9042DG1500). High pressure 60° core port. |
| 4 Weatherproof IP67 enclosure                                                     |                                                                                              |

### Technical Specifications

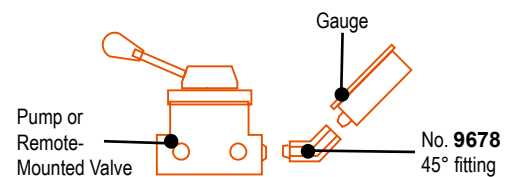
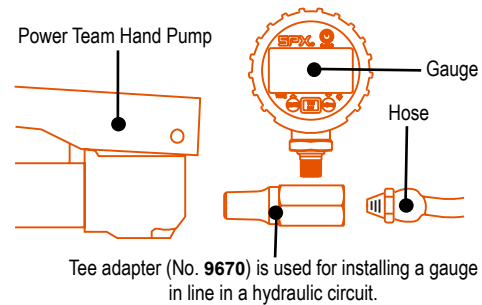


### Ordering Information

| Order No.  | Face Dia. (mm) | Rated Pressure (bar.) | Temp Range (°C) | Tool Use | IP Rating | Batteries Included | Battery Life (hrs.) | F.S. Accuracy | Prod. Weight (kg) |
|------------|----------------|-----------------------|-----------------|----------|-----------|--------------------|---------------------|---------------|-------------------|
| 9042DG     | 63,5           | 0-700                 | -20 to +60      | All      | IP67      | 2 x AA (LR6)       | 2000                | 0.50%         | 0,24              |
| 9042DG1500 | 63,5           | 0-1500                | -20 to +60      | All      | IP67      | 2 x AA (LR6)       | 2000                | 0.25%         | 0,24              |



### Typical Mounting Setups for 9042DG



### Hardware Requirements for 9042DG



- Does your pump have a 3/8" or 1/4" NPTF port?
- Are you direct-plumbing the gauge to your tool or seeking to quick-disconnect?
- Refer to the Power Team Catalog under the accessories section for a comprehensive range of fittings and couplers to meet your needs.



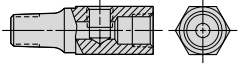
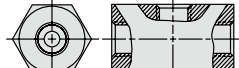
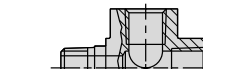
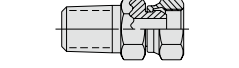
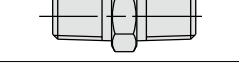
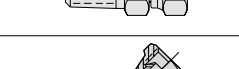
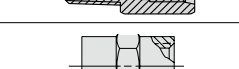
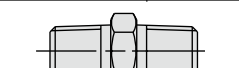
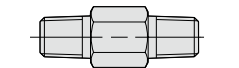
### Custom Scale

User programmable unit of measure allows you to set a custom scale to have the gauge display torque, force or weight to match the cylinder or tool being used.

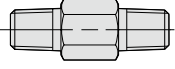
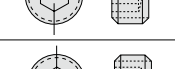
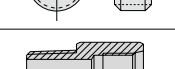
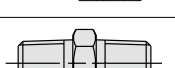
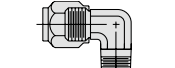
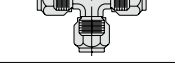
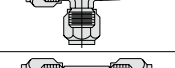
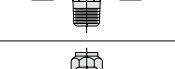
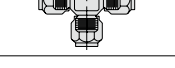
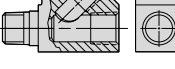
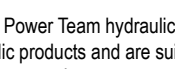


### Safety Tip

When using couplers to detach the gauge, always use a metal cover (ordered separately) on the pump side (female) coupler when the gauge is removed.

|                                                                                     | Order No. | Description                                                                                                                                  |
|-------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
|    | 9190      | <b>Hydraulic tubing.</b> 3/8" O.D. x .065" wall, 15,3 m. (10 pieces 1,53 m long.) Wt. 5,5 kg.                                                |
|    | 9670      | <b>Tee adapter.</b> For installing gauge between pump and hose coupling. Has 1/4" and 3/8" NPTF female and 3/8" NPTF male ports. Wt. 0,2 kg. |
|    | 9671      | <b>Double tee adapter.</b> Permits use of more than one cylinder in series with one pump. Three 3/8" NPTF female ports. Wt. 0,5 kg.          |
|    | 9672      | <b>Service tee.</b> Two 3/8" NPTF female internal, one 3/8" NPTF male external. Wt. 0,3 kg.                                                  |
|    | 9673*     | <b>Swivel connector.</b> 3/8" NPSM male, 1/4" NPSM female. Wt. 0,1 kg.                                                                       |
|    | 9674      | <b>Male connector.</b> 43 mm long, 1/4" x 3/8" NPTF. Wt. 0,1 kg.                                                                             |
|   | 9675*     | <b>Swivel connector.</b> 3/8" NPTF male, 3/8" NPSM female. Wt. 0,1 kg.                                                                       |
|  | 9676*     | <b>Swivel connector.</b> 1/4" NPTF male, 3/8" NPSM female. Wt. 0,1 kg.                                                                       |
|  | 9677*     | <b>45° swivel connector.</b> 3/8" NPTF male, 3/8" NPSM female. Wt. 0,1 kg.                                                                   |
|  | 9678      | <b>45° fitting.</b> Used when mounting gauge at an angle on connection such as 9670. Male and female 1/4" NPTF ends. Wt. 0,1 kg.             |
|  | 9679      | <b>Connector.</b> 1/4" NPTF female and 3/8" NPTF male. Wt. 0,1 kg.                                                                           |
|  | 9680      | <b>Coupling.</b> Both ends 3/8" NPTF female. Wt. 0,1 kg.                                                                                     |
|  | 9681      | <b>Street elbow.</b> Male and female 3/8" NPTF ends. Wt. 0,1 kg.                                                                             |
|  | 9682      | <b>Male connector.</b> 43 mm long, 3/8" NPTF male ends. Wt. 0,1 kg.                                                                          |
|  | 9683      | <b>Male connector.</b> 57 mm long, 3/8" NPTF male ends. Wt. 0,1 kg.                                                                          |

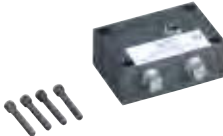
\* **CAUTION:** On part numbers 9673, 9675, 9676 and 9677 the female swivel end of these adapters is a straight pipe thread (NPSM) with a 30° seat. All male pipe fittings that are used with these female swivel adapters must have an internal 30° seat in order to effect a proper seal. All Power Team male fittings are manufactured with a 30° seat except 9687 and 9688.

|                                                                                      | Order No. | Description                                                                                                                                          |
|--------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | 9684      | <b>Male connector.</b> 57 mm long, 1/4" NPTF male ends. Wt. 0,1 kg.                                                                                  |
|    | 9685      | <b>Coupling.</b> 1/4" NPTF female and 3/8" NPTF female. Wt. 0,1 kg.                                                                                  |
|    | 9686      | <b>90° elbow.</b> 3/8" NPTF female ends. Wt. 0,2 kg.                                                                                                 |
|    | 9687      | <b>Pipe plug.</b> Heat-treated, 3/8" NPTF. Wt. 0,1 kg.                                                                                               |
|    | 9688      | <b>Pipe plug.</b> Heat-treated, 1/4" NPTF. Wt. 0,1 kg.                                                                                               |
|    | 9689      | <b>Connector.</b> 1/4" NPTF male and 3/8" NPTF female. Wt. 0,1 kg.                                                                                   |
|   | 9690      | <b>Male connector.</b> 43 mm long, 1/4" NPTF male ends. Wt. 0,1 kg.                                                                                  |
|  | 9692      | <b>Straight connector.</b> 3/8" tube x 3/8" male NPTF. Wt. 0,1 kg.                                                                                   |
|  | 9693      | <b>90° elbow.</b> 3/8" tube x 3/8" male NPTF. Wt. 0,1 kg.                                                                                            |
|  | 9694      | <b>45° elbow.</b> 3/8" tube x 1/4" male NPTF. Wt. 0,1 kg.                                                                                            |
|  | 9695      | <b>Tee.</b> 3/8" tube. Wt. 0,1 kg.                                                                                                                   |
|  | 9696      | <b>Male run tee.</b> 3/8" tube x 1/4" male NPTF. Wt. 0,1 kg.                                                                                         |
|  | 9697      | <b>Male branch tee.</b> 3/8" tube x 1/4" male NPTF. Wt. 0,1 kg.                                                                                      |
|  | 9698      | <b>Cross.</b> 3/8" tube. Wt. 0,2 kg.                                                                                                                 |
|  | 9699      | <b>45° gauge fitting.</b> 3/8" NPTF male and female, and 1/4" NPTF female at 45°. Wt. 0,3 kg.                                                        |
|  | 9705      | <b>Fitting, swivel.</b> 3/8" NPTF male to 3/8" NPTF female. 90° fitting with internal 370 micron screen. May be rotated 360° about male thread axis. |

**NOTE:** Power Team hydraulic fittings are intended for use with our high pressure hydraulic products and are suitable for use at max. working pressures of 700 Bar unless otherwise noted.

## MANIFOLD BLOCKS

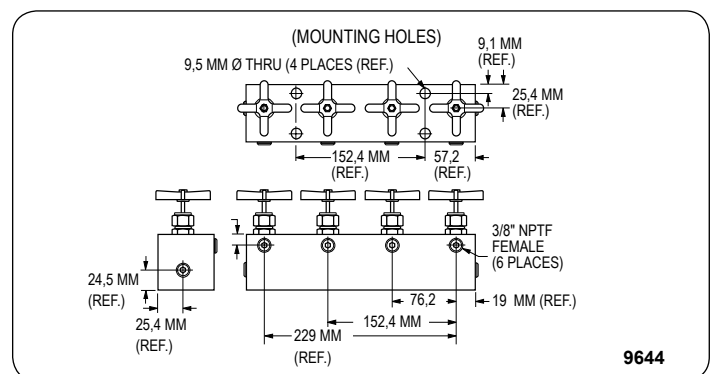
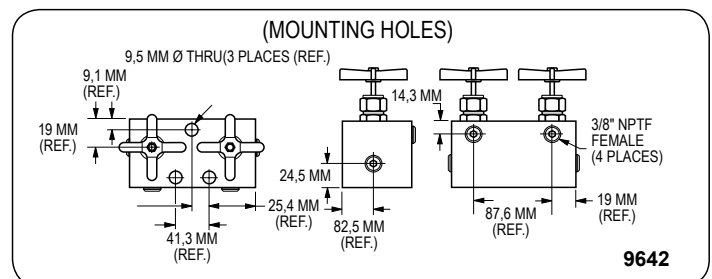
Select from a wide assortment of high pressure manifold blocks to customize your system.

|                                                                                     | Order No. | Description                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | 9691      | <b>"Y" Manifold.</b> Extremely useful when connecting two hydraulic cylinders to a single line. Has three 3/8" NPTF ports. Wt. 0,45 kg.                                                                                                                                                                                                                                                                           |
|    | 9634      | <b>Manifold block.</b> This manifold is for multiple-cylinder installations, has four 3/8" NPTF ports and two 6,4 mm mounting holes. 2" Square 1.5" thick. Wt. 0,7 kg.                                                                                                                                                                                                                                            |
|    | 9635      | <b>Manifold block.</b> This hex-shaped manifold offers extra versatility with six 3/8" NPTF ports and two 6,4 mm mounting holes. 2.5" Hex x 1.25 thick. Wt. 0,9 kg.                                                                                                                                                                                                                                               |
|    | 9617      | <b>Manifold block.</b> When a multiple-cylinder installation is required, this manifold is invaluable. Has six 3/8" NPTF ports to handle larger multiple-cylinder systems. Wt. 1,4 kg.                                                                                                                                                                                                                            |
|   | 9648      | <b>Manifold block.</b> This 178 mm long manifold block has seven 3/8" NPTF ports and two 6,4 mm mounting holes. Wt. 1,2 kg.                                                                                                                                                                                                                                                                                       |
|  | 9627      | <b>Manifold block.</b> This 406 mm long manifold block allows you to mount the 9575 or 9596 valves without interference. Has seven 3/8" NPTF ports and two 6,4 mm mounting holes. Wt. 2,7 kg.                                                                                                                                                                                                                     |
|  | 9626      | <b>Pump-mounted, manifold block.</b> Converts pump-mounted valves for use with remote-mounted valves. This manifold block is sub-plate mounted on the pump cover plate and provides 3/8" NPTF pressure and return ports. Maximum recommended flow rate is 19 l/min.<br><b>Note:</b> If used on PE30 or PG30 series pump, 12,7 mm longer mounting screws are required. Order four (4) No. 11956 screws separately. |

## MANIFOLD BLOCKS WITH NEEDLE VALVES

For independent multiple-cylinder operation, feature needle valves for precise manual control.

|                                                                                    | Order No. | Description                                                                                                                  |
|------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|
|   | 9642      | <b>2-Port Manifold.</b> Manifold with two needle valves for control of two cylinders. Has four 3/8" NPTF ports. Wt. 3,7 kg.  |
|  | 9644      | <b>2-Port Manifold.</b> Manifold with four needle valves for control of four cylinders. Has six 3/8" NPTF ports. Wt. 7,3 kg. |



## Technical Dimensions

| Manifold Number | A (mm) | B (mm) | C (mm) |
|-----------------|--------|--------|--------|
| 9617            | 177,8  | 38,1   | 38,1   |
| 9627            | 406,4  | 114,3  | 38,1   |
| 9648            | 177,8  | 38,1   | 38,1   |

